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School's out... forever: A study of juvenile crime, at-risk youths and teacher strikes ☆

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Abstract

The goal of this paper is to better understand the implications of forcing children to be in school, in terms of juvenile criminal activity. By combining a unique measure of unexpected school closings (using teacher strikes) with highly detailed criminal data, I show: (1) that incapacitation deters property crime and leads to increased violent crime to degrees that are potentially much larger than previously estimated, (2) that highly urban school districts are the most affected by incapacitation, (3) that juveniles with differing criminal histories are differentially impacted by incapacitation. I also confirm that these changes in crime rates reflect a change in overall crime and not a displacement of existing crime.
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1. Introduction

Although we have taken important strides in understanding the economics surrounding crime, there remains a great deal which we have yet to fully understand. The focus of this paper is to address the nature of juvenile crime in particular. Because juvenile crime can be especially hard to study, due to general limitations on access to data, economists still have a lot to explore on this front.

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Over the past 20 years, economists and social scientists have attacked the problem of crime in four distinct ways. Specifically they have analyzed the potential effects of deterrence, retribution, rehabilitation and incapacitation as forces for reducing crime (Ehrlich [4]).[1] Deterrence, a widely explored topic, stresses the importance of imposing penalties as an effective means of preventing crime, because the perceived costs to criminals of criminal activity increases (Mocan and Rees [14], Levitt [11], Freeman [6], Ehrlich [5], Lochner [13]). Retribution addresses the actual punishment of criminals. It suggests that a criminal experiences an increased cost to his crime when a punishment is imposed on him and/or other criminals (Levitt [11]). Rehabilitation stresses the importance of reforming criminals, through treatment and rehabilitative programs, in preventing future crime (Cueller et al. [3]). Finally, incapacitation deals with the notion that the physical lock-up and detaining of criminals may be an effective means of preventing crime (Freeman [6], Lochner [12]). This paper will closely examine the role of incapacitation in preventing juvenile crime.

Specifically I deal with the effect of incapacitating juveniles in school on juvenile crime. Brian Jacob and Lars Lefgren [10] were the first to address this topic. They examine the effect of school attendance on juvenile crime by using teacher in-service days, days where teaching professionals are required to attend work and children are not required to be at school, as a source of variation in student attendance. While their use of in-service days is very clever, it is a somewhat limited tool for examining the effect of incapacitation. To overcome these limitations, I use teacher strikes as a source of variation of student school attendance.

One primary advantage to using teacher strikes over in-service days as a source of variation in school attendance is that strikes often occur in blocks larger than one individual day. This allows me to observe the effect of prolonged absence from school, and to see how juvenile crime trends over time. If juvenile criminal habits do change over time, then a one day absence cannot reflect the average effect of school incapacitation.

Another advantage of utilizing teacher strikes is that they are relatively unpredictable. Since in-service days are planned at the beginning of the regular school year, parents have adequate time to make arrangements for their children during these off-days. In contrast, school closures due to teacher strikes are often reported in local newspapers only a couple of days before they are likely to take place, and even then, there is no guarantee that these reported strikes will actually occur. Additionally, a parent's ability to plan for their children's activities during strike days is complicated by the fact that details revealed in newspapers and other information sources may prove to be inaccurate or unjustified.[2] The late-breaking and incomplete nature of information leaves parents little time to plan their children's activities during the days of a teacher strike. Thus teacher strike days are likely to result in more unsupervised students, and estimates that provide an upper bound of the possible effects of incapacitation.

The main drawback to this use of strike as variation is that teacher strikes are often very sparse, and mostly occur in a only a few school districts. In order to overcome this difficulty, I rely on data from Washington State. Washington provides an ideal environment for this type of study

[1] I am not arguing that these four ideas are mutually exclusive ways of preventing crime.

[2] For instance, a school district may report that it is sure teachers will return to work when a court injunction is issued, and subsequently the teachers may defy the injunction.

because it has an extensive strike history as well as a detailed juvenile arrest data set that dates back as far as 1980.[3]

2. The identification strategy

As stated earlier, the objective of this analysis is to measure the impact of school incapacitation on juvenile crime rates by using teacher strike days as variation in student absence from school on an ordinary school day. Arguably, teacher strikes are a source of variation that is exogenous to variations in juvenile crime. Issues that force teachers to strike include pay, class size and teacher planning time, none of which are likely to influence daily variations in juvenile crime.[4] Also it is worth noting that due to the limitations of the data, my unit of observation takes place at the zip code level, though I will talk more about this in a moment.

In order to draw direct comparisons between crime on strike vs. non-strike days, I must be able to control for days in which schools are closed for other reasons. This would include days that coincide with spring, summer and winter vacations, weekends, teacher in-service and half-days, national holidays, etc. However since the data cover a large span of time (22 years), and since each Washington school district determines their own school calendar, it is impossible to know where all of these non-typical school days occur.[5] Further, a prolonged teacher strike sometimes results in a reworking of the existing school calendar (largely unobservable) so that a school district can meet the requirement set by the state that the school year last for 180 days.[6] Since I cannot control for these non-typical school days with complete of accuracy, I must include only those days which I am strongly confident students are regularly scheduled to be in school.

To minimize any possibility that non-typical school days could bias the results, I take extra measures when eliminating questionable days. I begin by eliminating weekends and national holidays.[7] If a national holiday falls on a Saturday or Sunday, I eliminate the preceding Friday or subsequent Monday, respectively. The first two Fridays of October are dropped because they are both traditional days for the state mandated teacher in-service day. The entire months of June and July were also excluded, and only the last three school days in August are allowed in the sample. Finally I eliminate the first week in January, the last two full weeks (at least) of March, the first two weeks of April (at least 8 school days), and the last two full weeks (at least) of December. Based on these criteria, I can be reasonably certain that each observation in the analysis is a

[3] Unlike most states in the US, teacher strike activity in Washington remains very much active to this day. The most recent strike event took place in 2003 in Marysville school district. This strike lasted nearly two months (approximately 50 total days). A summary table describing teacher strikes in Washington can be seen in Appendix A.

[4] In fact, areas where juvenile crime may be an increasing problem are arguably less-likely to see a teacher strike, because increasing crime has been shown to be positively correlated with higher teacher pay (Grogger [9]).

[5] Each school district may vary on the scheduling of breaks, half-days, inservice days, etc. To the best of my knowledge, there is no state or federal agency that has archived these calendars over the 21 years. Since the advent of increased technological integration with school systems, some districts are beginning to electronically archive these calendars though at present these calendars generally only go back a couple of years. The Washington State Superintendent of Public Instruction was able supply me with sufficient calendars for most districts, for each year from August 2000 to June 2004.

[6] The shortening of spring and winter breaks is common, as well as the lengthening of the school year into summer vacation.

[7] Although Thanksgiving falls on Thursday, both Thursday and Friday are dropped from the sample because students are given both of those days off from school.

typical school day.[8] Days which are included in the sample but are not ordinary school days will serve to downwardly bias the results, because more juvenile crime should be naturally occurring on those days. This is only true because this measurement of strikes as a treatment is limited to the days where students have missed an expected day of school.[9]

2.1. Zip code matching

My data set consists of individual arrest data, where each juvenile is matched to his home address zip code. Since zip codes are set by the Address Information division of the US Post Office, they look very different from other boundaries like county borders, congressional districts, and most importantly school districts. In order to match each juvenile (and his/her arrest) to his/her appropriate school district, I must address two problems. The first problem is that zip codes often spill over multiple school districts. Therefore it is often uncertain which school district a juvenile is in, given their home zip code. The second problem is that zip codes are redefined over time. First I will focus on how zip codes change, and how this problem is resolved so that identification strategy is preserved.

As cities and towns develop both inside and outside city limits, the Post Office creates new zip codes and new zip code offices to handle increasing mail traffic. When these new zip codes are created, they are drawn strictly as subsets of existing zip codes.[10] The creation of these new zip codes makes zip code fixed effects useless from one year to the next. In addition, it is also very difficult to pinpoint when new zip codes are created. However it is possible to see which zip codes were created and what pre-existing zip code they were originally a part of. By using zip code maps of Washington State from 1984 and 2000, I am able to map each subset zip code back into its original (1984) zip code.[11] Essentially I treat zip codes in Washington as if they are never partitioned, beginning in 1984. This method leaves me with 510 zip codes in total and a stable zip code definition over a 22 year time frame.

The second problem is matching these zip codes to their corresponding school district. Zip code overlap with school districts make this matching process difficult; however, the approach I take is sound and fairly straight forward. I began by totaling the number of schools in each zip code, noting which district they serve.[12] I then used the total number of schools in that zip code for a given district and divided it by the total number of schools in that zip code, regardless of their district. Essentially what I am left with is a probability, which I will call p^*, which reflects the probability that a child living in a particular zip code is also part of a particular school district. This p^* measure allows me to assign a proper strike treatment to zip codes.

There are several features of zip codes that simplify this matching process. One helpful factor is that many zip codes do not overlap school district boundaries at all. This leaves out any guess

[8] This is not a perfect matching process because half days and minor differences between districts are impossible to pinpoint.

[9] The timing of strikes is very particular and seldom overlaps with non-school days, except weekends.

[10] So far I have not been able to find an instance in Washington State since 1984 where a zip code was created from two or more existing zip codes.

[11] Four maps of Washington State zip codes were provided by the Western Economics Research Co., Inc., courtesy of the Suzzallo Library at the Washington Library. They encompass they following years: 1984, 1989, 1992 and 1994. Zip code maps of Washington State for the year 2000 are made available by the Census.

[12] I did this using a master list of schools (as of the 2003–2004 school year) that includes each school's address, grade level served, school code and district name. This list was provided by the Office of Superintendent of Public Instruction in Washington State. It is also publicly available.

work. Another helpful point is that zip codes and school districts often share a significant number of common boundaries. These shared boundaries occur where divisions seem logical, such as along a county boundary, or a major river or highway. These convenient features lessen the confusion of the matching process. It is also worth noting that not all zip codes contain schools.[13] In my sample there were four zip codes without any schools; two were in the heart of Seattle, and two were located in rural areas. Conveniently, none of these four zip codes overlapped school district boundaries, so assigning them to the appropriate school district was simple.

The last concern I am left with deals with how *school districts* change over time, if at all. If school district boundaries are changing frequently and dramatically over time, then the matching process breaks down. However, what I find is that school district boundaries are stable over time. In order for a district to change its boundaries, it must engage in costly and time-consuming legal processes; however, that does not imply that such adjustments never occur.[14] In fact, the two most frequent changes to school district boundaries seem to be district consolidation and district dissolution.[15] Neither changes turn out not to be problematic. These changes only occur a total of four times over the course of the 22 years and involve districts which do not experience any teacher strikes. I treat these integrated districts as if they had never been separate.

The school district-zip code matching process I invoke does not provide perfectly accurate matches in all cases, but I can be certain of how any mismatching will bias the results. If I say there was a strike in an unaffected zip code, the effect of the strike will be biased toward zero because juvenile crime should be unchanged. If I say that there was not a strike in an affected group, then the strike effect will fail to pick up any increase in juvenile crime rates. Clearly any mismatching that arises from the imperfect matching process will downwardly bias the results.

2.2. The basic model

In this basic model I am attempting to explain changes in juvenile crime for ordinary school days as a function of teacher strikes. I can start by expressing a simple regression model in the following form:

$$\text{Juvenile School Day Crime} = \alpha + \beta(p^*)(\text{Teacher Strike}).$$

Again, all of the information from the crime data is reported at the zip code level; however, the teacher strikes occur at the school district level, so I must adjust. Here p^* again represents the probability that the student population of a zip code is treated by the teacher strike. Rather than have fixed effects to take into account differences across zip codes, I want to initially include specific zip code characteristics to make sure the data are well-behaved. I consider income levels, welfare status, parental education, juvenile work status, juvenile gender, single parent households and community characteristics in this model. Also I need time fixed effects to control for temporal changes over 22 years. I therefore include year, month and day fixed effects into the model. The variable Year^i is a dummy variable for year i, which takes a value of 1 when $y = i$ and zero

[13] The zip codes I use in the sample are zip codes with residences. Many zip codes like business zips, P.O. Box zips, and other non-residential zips, do not contain schools. These kinds of zips are not included in the data sample.

[14] For an outline of regulations and requirements surrounding district boundary changes, please refer to the publication, "Changing School District Boundaries: A Lay Person's Guide" published by the Washington State Board of Education in conjunction with the Office of Superintendent of Public Instruction.

[15] Consolidation is when two or more districts form a new super-district. Dissolution is when a one or more districts are absorbed into existing districts.

otherwise. Similarly Month^j and Day^k are dummy variables for month j and day k. Each takes on a value of 1 when $m = j$ and $d = k$ respectively and zero otherwise. The regression model now takes the form:

$$\begin{aligned}\text{Juvenile School Day Crime}_{myd} &= \alpha + \beta_1(p^*)(\text{Teacher Strike})_{mydz} \\ &+ \cdots + \beta_2(\text{Median Income})_{mydz} + \beta_3(\text{Welfare})_{mydz} + \beta_4(\text{Parent Education})_{mydz} \\ &+ \cdots + \beta_5(\text{Student Employment})_{mydz} + \beta_6(\text{Juvenile Gender})_{mydz} \\ &+ \cdots + \beta_7(\text{Urban})_{mydz} + \beta_8(\text{Single Parent})_{mydz} + \sum_i \delta_i(\text{Year})^i + \sum_j \delta_j(\text{Month})^j \\ &+ \cdots + \sum_k \delta_k(\text{Day})^k.\end{aligned}$$

Once I am satisfied that the data exhibit all of the normal properties one would expect from specifying a model of juvenile crime, I can then shift the analysis to include zip code fixed effects take into account differences which I cannot control for. Here Zip^n is a dummy variable for zip code n, which takes a value of 1 when $z = n$ and zero otherwise. When I move to this specification, the final regression model looks like the following:

$$\begin{aligned}\text{Juvenile School Day Crime}_{mydz} &= \alpha + \beta_1(p^*)(\text{Teacher Strike})_{mydz} \\ &+ \cdots + \sum_i \delta_i(\text{Year})^i + \sum_j \delta_j(\text{Month})^j + \sum_k \delta_k(\text{Day})^k + \sum_n \delta_n(\text{Zip})^n.\end{aligned}$$

3. The data for this model

My juvenile arrest data comes from "Washington Juvenile Court Case Records" made available by the National Juvenile Court Data Archive. It provides juvenile arrest data over 22 years, spanning from 1980 to 2001. This data set is both lengthy as well as highly detailed at the individual level. It provides the home address zip codes of the offenders, the nature of the crimes committed, the dates of the offenses (as well as the arrests), as well as many other important characteristics surrounding the reported arrests. Since individual crime data is reported at the zip code level, the zip code becomes my unit of observation, where school districts are matched to their zip codes according to the process described earlier. In total when I include only Washington juveniles who were arrested for crimes that occurred during the ordinary school days pre-selected for this study, I have 401,864 arrest cases starting in 1980.[16] I report some descriptive statistics of the juveniles who make up this data set in Appendix B.

In addition to the juvenile arrest data for Washington State, I use annual data provided by the "Census 2000 Summary File 3" and "1990 Summary Tape File 3" to derive zip code characteristics. These Census files give zip code level information on population and school enrollment numbers, as well adult educational attainment, the number of households who are welfare recipients, type of community, median household income and student employment.

[16] Although some conviction data is contained within the data set, these data are not used to measure criminal activity. The primary reason for this is due to the fact that the juvenile justice system works quite differently from adult court, and conviction status is difficult to determine. For juveniles, adjudication (conviction) status is only determined by the courts when an offense is petitioned, an action which often does not occur.

Finally, teacher strike information came from the report "Public Employee Strikes in Washington" published in 2003 by the Public Employees Relations Commission in Washington State. It covers all Washington public employee strikes since 1967[17]; however, since this publication lacks some important details about these strikes, it has been supplemented with specific information from news articles provided by the Associated Press and the Seattle Times to strengthen the accuracy of the treatment effect.

3.1. The methodology

Choosing a proper methodology for this analysis requires some extra care. Using a standard OLS approach is not appropriate because this analysis utilizes daily count data, with zero makes up a large percentage of the observations for any given day. In addition the distribution of the dependent variable is also skewed towards zero. For count data such as this it would normally be appropriate to use a Poisson estimation; however, the fit for a Poisson regression for this data ends up being poor.[18] Specifically the Poisson estimation does not account for strangely behaved standard errors and overdispersion that exists in the data.

In this case, overdispersion may take the form of a single juvenile committing several crimes on a single strike day, or a gang of juveniles committing what could be considered one crime. Therefore I use negative binomial regression analysis which deals with problems of overdispersion that a Poisson does not correct for.[19] By using daily total crime in a zip code as my dependent variable with total student population in said zip code as my exposure, the negative binomial model allows me to estimate the linear relationship between strike days and the log of the mean number of crimes. Thus the estimated strike coefficient represents an approximation of the percentage change in total crime on strike days.[20]

3.2. Defining the parameters

As described in the model, I begin by controlling for differences between zip codes directly rather than using zip code fixed effects. Each variable measurement is derived from data taken from the 1990 and 2000 Census. Because these characteristics are only available for these two periods (1990 and 2000) I interpolated the values of each characteristic over the 22 years using

[17] Since enactment of Public Employees Collective Bargaining Act in 1967.

[18] A preliminary test of the data shows that the dependent variable generally has a standard deviation roughly 3 times larger than the mean. This suggests that overdispersion exists in the data. In addition, I observe large values of the Chi squared terms in the Poisson "goodness of fit" tests. Both of these characteristics suggest that the Poisson model in not the correct model for these regressions.

[19] The overdispersion parameter alpha is shown to be significantly different from zero. Thus I can conclude that the negative binomial regression significantly different from the Poisson regression. In addition, the qualities of my data and my choice of model are consistent with Jacob and Lefgren's choice of model.

[20] The reader should be aware that the percentage effects of strike days that I report in the text of the paper represent the average percentage effect estimation of the strike coefficient across all specifications types I–V. However in the tables only the coefficient for the type V specification is reported. In general the reported strike coefficient in each table (with type V specification only) is the top range of the estimated effect. Therefore the percentage effects reported in the text can be considered conservative estimates of the effect of strike days on crime for various types and groups.

Table 1
Descriptive statistics of covariates

	Full sample		Sub-samples					
	All zip codes		Urban		Suburban		Rural	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Specification I								
Strike	0.0012	0.0339	0.0035	0.0574	0.0006	0.0246	0.0003	0.0181
Median income	18,860.91	6535.39	21,425.87	6515.71	17,143.14	3680.42	18,024.51	6567.65
Welfare	0.0787	0.0705	0.0701	0.0611	0.0887	0.0584	0.0819	0.0754
Urban	0.2582	0.4190	0.9416	0.1115	0.0290	0.1374	0.0102	0.0678
Poor parental educ.	0.1465	0.1102	0.1209	0.0905	0.1300	0.0636	0.1592	0.1181
Juvenile maleness	0.5273	0.0949	0.5063	0.0866	0.5136	0.0214	0.5377	0.1204
Student employment	0.2906	0.2126	0.3710	0.1709	0.3233	0.1557	0.2539	0.2256
Single parent house	0.0829	0.0920	0.0890	0.0734	0.0973	0.0644	0.0780	0.1015
Total students	1554.31	2167.97	3792.82	2644.76	2183.10	1626.19	542.45	878.20
Specification IV								
Strike	0.0035	0.0580	0.0106	0.0996	0.0016	0.0394	0.0008	0.0287
Median income	18,763.85	5879.70	21,457	6418.83	17,213.68	3743.37	17,786.28	5500.05
Welfare	0.0682	0.0567	0.0615	0.0503	0.0759	0.0465	0.0707	0.0605
Urban	0.2610	0.4205	0.9414	0.1109	0.0339	0.1474	0.0076	0.0569
Poor parental educ.	0.1339	0.0853	0.1095	0.0668	0.1220	0.0517	0.1461	0.0929
Juvenile maleness	0.5220	0.0941	0.5050	0.0704	0.5134	0.0273	0.5300	0.1238
Student employment	0.2790	0.1931	0.3575	0.1526	0.3115	0.1309	0.2405	0.2057
Single parent house	0.0846	0.0743	0.0940	0.0575	0.1022	0.0537	0.0781	0.0824
Total students	1621.75	2247.55	3967.51	2701.42	2256.79	1646.81	525.28	851.76

Note. Urban, Suburban and Rural sub-samples limit the sample to zip codes where 51% or more of the population are considered (according to the Census) to be living in an Urban, Suburban or Rural area.

an exponential function rather than a linear function to describe the path of the trend.[21] I started by solving for the growth rate ($G_{i,zip}$) of each variable i_{zip}:

$$G_{i,zip} = \left((\text{Census}_{i,zip\ 2000}/\text{Census}_{i,zip\ 1990})^{1/10}\right) - 1.$$

Then I interpolated each year's observation according to the equation:

$$\text{Variable } i_{zip} \text{ in year } t = \text{Census}_{i,zip\ 1990} * (1 + G_{i,zip})^{n}, \quad \text{where } n = t - 1990.$$

Overall this analysis uses 7 variables to capture differences in communities. Descriptive statistics for these variables can be viewed on Table 1.

Income is measured using Median Household Income, which comes directly from the Census data. I then convert this measure of income to real median income by controlling for cost of living across the 22 year period. I divide median income by the consumer price index where 1980 is the base year. In other words, median income for every year is expressed in terms of the value of dollars in 1980. The Welfare variable is a percentage of households in a zip code that currently receive public assistance income. Of course I expect that local poverty is a serious factor in juvenile crime, so both of these variables seem to be relevant (Mocan and Rees [14]). Juveniles

[21] I have also interpolated the value of each characteristic along a linear path; however, when I do this I end up with some numbers that are not feasible do to the imprecise nature of this process. When I impose minimum and maximum constraints on these variables, I find that defining these variables linearly or exponentially does not significantly change the results.

from poor neighborhoods are more likely to be less educated, and therefore more likely to be criminally active overall (Freeman [7]). Since juveniles from poor households have a smaller opportunity cost of committing crime, I expect Median Income to be negatively correlated with juvenile crime, and naturally the converse should be true for Welfare.

The Parental Education variable is defined as the percent of adults (25 years+) who have not completed the equivalent of a high school education. My prediction here is that as adult dropout rates increase, juvenile crime increases. Because poor adult/parental education equates to less adult human capital, parents are likely to have lower demand for child quality (Becker and Tomes [1,2]). Therefore juveniles in these neighborhoods should also have a lower opportunity cost of crime. In addition, one can argue that poor adult/parental education may also imply that parents have fewer resources to provide monitoring for juveniles, less overall time to provide supervision of their own children (parents may be working long hours or more than one job), or possibly uninformed/undesirable parenting techniques.

Our Student Employment variable describes the percent of 16–19 year olds who are both employed and also enrolled in school. The prediction of this variable is not clear. On one hand, students who are employed in after school jobs may develop a sense of responsibility from owning a job, or that they may fear losing future wages should they be caught committing a crime. This suggests that Student Employment should negatively influence juvenile crime. On the other hand, juveniles who work may have greater access to reliable transportation, or less restrictive parents, both of which could contribute to a greater propensity to commit crime. In this case, Student Employment would positively influence juvenile crime.

The Juvenile Male variable measures the percent of the juvenile population in a zip code who are male. Since juvenile males traditionally make up the majority of juvenile crime, juvenile crime for a zip code should increase as a juvenile population there becomes proportionately more male.[22]

The Single Parent variable is the percent of single parent households in that zip code. One expects that this variable should be positively correlated with juvenile crime so that more single parent households lead to more potentially unsupervised juveniles, and potentially greater juvenile crime. The Urban variable is the percent of the total zip code population who live in an urban community. This should pick up basic differences between community types; however, I also interact this variable with the strike measure to see if strikes have differential impacts in different types of communities. For several reasons, I expect crime in urban areas to differ from that of Rural or Suburban areas. Urban areas feature many characteristics, such as increased criminal opportunity, higher pecuniary benefit, and increased criminal anonymity, which makes criminal activity in these areas more desirable (Glaeser and Sacerdote [8]). As such this Urban variable should be positively correlated with juvenile crime. Lastly, the Strike variable is a dummy variable, with a value of 1 describing a strike event, and zero otherwise.

4. Regression analysis

The initial results of the negative binomial regression analysis seem to confirm previous findings about the nature of school incapacitation. Column 1 in Table 2 shows that the presence of

[22] From 1993 to 2002, males made up about 71–75% of total juvenile arrests. For more information about juvenile crime by gender, please reference the *Crime in the US* annual report published by the Federal Bureau of Investigation.

Table 2
Negative binomial regression onto total crime with full data set

Variables	Full data set				
	(I)	(II)	(III)	(IV)	(V)
Strike	**0.195****	**0.187****	**0.210****	**0.260****	**0.220****
	(4.06)**	**(3.86)****	**(4.16)****	**(5.06)****	**(4.38)****
Median income	−0.000058**	−0.000059**	−0.000059**	−0.000055**	–
	(102.47)**	(83.82)**	(60.76)**	(52.80)**	–
Welfare	0.475**	0.452**	0.520**	0.821**	–
	(7.71)**	(5.96)**	(4.97)**	(6.28)**	–
Urban	0.132**	0.112**	0.097**	0.055**	–
	(26.96)**	(18.45)**	(11.54)**	(6.14)**	–
Poor parental educ.	0.329**	0.346**	0.398**	0.482**	–
	(10.40)**	(8.88)**	(7.45)**	(6.67)**	–
Juvenile maleness	−0.330**	−0.341**	−0.355**	−0.396**	–
	(28.06)**	(26.24)**	(22.32)**	(22.72)**	–
Student employment	0.291**	0.297**	0.290**	0.296**	–
	(17.49)**	(14.43)**	(10.23)**	(9.16)**	–
Single parent house	0.500**	0.495**	0.466**	0.985**	–
	(9.52)**	(7.64)**	(5.20)**	(9.67)**	–
Alpha	0.989**	1.009**	0.997**	0.956**	0.759**
	(156.56)**	(126.04)**	(91.12)**	(87.74)**	(79.06)**
Number of obs.	1,703,910	1,128,630	583,440	479,400	479,400
Time fixed effects	Y	Y	Y	Y	Y
Zip fixed effects	N	N	N	N	Y
R-squared	0.0367	0.0364	0.0367	0.0279	0.0559

In columns I–V, the sample is re-specified as follows (Type I through Type V):
(I) – Sample includes all ordinary school days from 1980–2001;
(II) – January, March and May are excluded from the sample 1980–2001;
(III) – Includes only April, September and October from 1980–2001;
(IV) – Only April, September and October from 1984–2001;
(V) – Only April, September and October from 1984–2001, with zip code fixed effects.

Z-statistics are given in the parentheses for every table.

** Significant at the 1% level or smaller.

To interpret these regressions, I calculate the marginal effect of these variables and divide by the predicted value of the dependent variable at the mean of the covariates. The average marginal effect of the strike variable is roughly (0.0157), which represents a 21.4% increase in average total juvenile crime on strike days.

teacher strikes seems to have a positive and significant effect on juvenile criminal behavior. But before I begin to quantify the effect of school incapacitation on juvenile crime, I want to make sure that the data are behaving properly.

The signs of the covariates in column 1 of Table 2 seem to support the predictions made on the effects of income, welfare, parent's education and urban status with respect to juvenile crime. First, we see that the reported coefficients for Median Income and Welfare are negatively and positively significant, respectively. This implies that children in lower income households and welfare receiving households are more likely to engage in juvenile crime. These results support the predictions associated with poorer living conditions. In addition, lower parental education leads to higher crime as well. Again, for several reasons this result makes sense. The coefficient of the Urban variable is positive and significant, showing that urban communities experience more overall juvenile crime. The reported coefficient of the Student Employment variable is posi-

tive and statistically significant. This may imply after-school jobs provide students with greater resources which they may use to commit crime. Further, it may reflect the fact that as more students get after-school jobs, they are spending less time increasing their human capital (through study, after-school activities, etc.).

One problem is that the coefficient of the Juvenile Male variable is negative and significant, which contradicts the prediction. It says that across juvenile populations, those that are comprised of relatively more females, experience more juvenile crime. One possible explanation for this result is that it may be expressing problems in variable measurement for small communities. For rural communities there are two basic problems. The first is that with a small population of juveniles, the variation of the Juvenile Male variable increases significantly.[23] The second is that this unusual variation makes interpolating this variable equally unreliable. Since a significant portion of zip codes have very small populations of juveniles, it is possible that the negative result I observe is being driven by rural zip codes that have atypical juvenile gender characteristics.[24]

To check the sensitivity of the initial results I begin by eliminating months in which strikes did not occur from the sample: January, March, and May (Specification II).[25] I then went a step further and eliminated every month other than September, October and April from the sample (Specification III). September and October are preserved because they are the most common months for strikes, and April was included because in 1991 there was a "state-wide" teacher strike that involved at least 41 school districts. The results of Specifications II and III are reported in columns 2 and 3 of Table 2 respectively. These results are not significantly different from Specification I.

Recalling the zip code matching process described earlier in the paper, I am able to accurately match back to 1984 zip codes. However the crime data dates as far back as 1980. To deal with potential zip code mismatches before 1984, I drop the first four years of the sample (Specification IV).[26] This fourth specification shows the strike variable remains significant as the sample is modified (column 4 of Table 2). These results seem robust to many different representations of the data set.

It is also important to verify that these results are not being driven by zip code characteristics being specified inaccurately. The most immediate check of this is to use zip code fixed effects in lieu of explicit variables to capture the difference. Column 5 shows the addition of zip code fixed effects does not significantly alter the results.

The coefficient of the strike variable column 5 of Table 2 shows that strikes have a positive effect on juvenile crime that is statistically significant.[27] To quantify the impact of this increase, I take the partial derivative of the dependent variable with respect to the strike variable (∂Totalcrime/∂Strike). When I divide this marginal effect by the predicted value of the

[23] For zip codes with juvenile populations less than 500, the standard deviation of the Juvenile Male variable is approximately 3 times larger than the standard deviation of those zip codes with 500 or more juveniles.

[24] In fact when I look at only urban zip codes, I see that this variable seems to correct itself, which gives credence to this argument. Further, Appendix C shows that when the smallest zip codes (population less than 500 juveniles) are dropped from the sample, the negative significance of the coefficient is lost. Thus it seems plausible that poor interpolation of this variable for sparsely populated areas has tainted the coefficients.

[25] Over the 22-year period, there was one school district that orchestrated a strike in January, but it only lasted for one day.

[26] In doing this I eliminate 9 strike events, 3 of which lasted 9 or more days.

[27] I have specified a zip code fixed effects regression that includes all of the Census covariates for each Table 2 through 5. Though each regression is not explicitly reported on each table, the results are consistent with the overall findings on each table.

dependent variable evaluated at the mean of the strike variable, I am left with the percentage change in total crime on strike days. In the full data set, the marginal effect of the strike variable suggests that total juvenile crime increases by an average of 21.4% on days when strikes occur. This is more than a modest change in total crime.

4.1. Community differences

Since the full data set includes all communities, some of which may look and behave very differently from one another, I want to test whether juvenile crime in different communities is differentially affected on strike days. To capture community differences, I run separate regressions for each community type, where I restrict the sample to only those communities in which 51% or more of the population in a zip code live in one of three types of communities: urban, suburban or rural.[28] The results of these regressions can be viewed in the columns of Tables 3, 4 and 5. These results show that unexpected school absences significantly influence juvenile crime only in urban communities.[29]

This result is not at all unreasonable, given there are many differences between these types of communities that no doubt influence criminal behavior. It seems understandable that rural communities experience no change in juvenile crime when a strike occurs. Rural communities may provide less opportunity for crime, as well as a small town atmosphere that makes anonymity difficult. The lack of a significant effect from strikes in suburban communities is less believable since they arguably provide greater opportunity and anonymity; however, there may yet be other (possibly unobservable) characteristics about suburban characteristics that naturally deter juvenile crime on these strike days. For instance, suburban communities may be populated by more involved parents, more effective emergency resources, etc. The strike effects in urban communities however are positive and significant. In urban communities juvenile crime increases by an average of 17.7% on strike days.[30]

Again I run the same specification tests to test the robustness of these results. The signs of the covariates in every urban sub-sample are the same as in Table 2, except for Juvenile Male. The Juvenile Male covariate does take on a positive and significant coefficient in the urban subsample while in the suburban and rural sub-samples it remains negative and significant.

4.2. Differences in offense types

Since school incapacitation may be differentially affecting the types of crimes that juveniles are committing, I partition the dependent variable measure of total crime into 5 specific crime types. These types of crimes include: drug and alcohol related crimes, mischievous crimes, property crimes, violent crimes, and finally weapons and endangerment crimes. I still consider all

[28] The reader should be aware that this specification implies that zip codes which are considered one third urban, one third suburban, and one third rural, will therefore be excluded from this analysis. However because of the way in which the Census defines these communities, these kinds of zip codes make up a very small portion of the total sample.

[29] It is also true that the variation of daily total crime within zip codes generally differs across community types. For this data set the standard deviation of total crime within each zip code ranges from 0 to 2.52, with an average standard deviation of 0.074 in the 5 least populated zip codes, and an average standard deviation of 1.551 in the 5 most heavily populated zip codes.

[30] The huge decrease in the magnitude of the strike coefficient from the full sample to urban sub-sample indicates that small populations in rural communities are exaggerating the effects of the strikes in the full sample.

Table 3
Negative binomial regression onto total crime with urban sub-sample

Variables	Urban sub-sample				
	(I)	(II)	(III)	(IV)	(V)
Strike	**0.151****	**0.129***	**0.178****	**0.211****	**0.214****
	(3.07)**	**(2.60)***	**(3.50)****	**(4.11)****	**(4.26)****
Median income	−0.000061**	−0.000062**	−0.000062**	−0.000057**	–
	(83.07)**	(67.39)**	(48.42)**	(42.17)**	–
Welfare	1.394**	1.369**	1.418**	1.703**	–
	(16.40)**	(13.01)**	(9.79)**	(9.46)**	–
Poor parental educ.	0.398**	0.403**	0.449**	0.462**	–
	(9.49)**	(7.79)**	(6.31)**	(4.65)**	–
Juvenile maleness	0.602**	0.577**	0.482**	0.397**	–
	(11.13)**	(8.60)**	(5.16)**	(3.62)**	–
Student employment	0.214**	0.224**	0.214**	0.157**	–
	(9.10)**	(7.67)**	(5.34)**	(3.47)**	–
Single parent house	0.739**	0.790**	0.945**	1.276**	–
	(8.28)**	(7.12)**	(6.24)**	(7.55)**	–
Alpha	0.676**	0.685**	0.666**	0.638**	0.494**
	(114.06)**	(91.00)**	(65.18)**	(62.65)**	(54.87)**
Number of obs.	454,953	301,312	155,601	129,593	129,593
Time fixed effects	Y	Y	Y	Y	Y
Zip fixed effects	N	N	N	N	Y
R-squared	0.0484	0.0484	0.0487	0.0400	0.0635

Note. Table 3 limits the sample to include only zip codes where 51% or more of the population lives in an area considered by the Census to be Urban.

In columns I–V, the sample is re-specified (Type I through Type V) as in Table 2.

* Significant at the 5% level.

** Idem., 1% or smaller.

To interpret these regressions, I have to calculate the marginal effect of these variables. The marginal effect of the strike variable is 0.121, which represents a 19.68% increase in average total juvenile crime on strike days.

three community classifications when doing this analysis because I may yet find significant results in types of crimes for rural and/or suburban communities.[31] Table 6 shows the results of this extended analysis.

Rural and suburban communities continue to experience no change in juvenile crime resulting from strike days regardless of the types of crime that is being committed. Urban communities though do experience an array of changes in juvenile criminal behavior. Both mischievous crimes and property crimes seem to increase in the presence of a teacher strike. I estimate that mischievous crime increases by 48% on average for days when strikes occur.[32] Property crime increases by an average of 28.8%. This change in property crime is nearly double previous estimates in the literature. Violent crime decreases by approximately 31.5%.[33] Drug and

[31] This could arise if increases in certain crimes were offsetting decreases in other types of crimes.

[32] It is difficult to compare my measure of mischievous crime to Jacob and Lefgren's measure of minor offenses. They define minor offenses as "NIBRS Group B offenses"; however, my measure of mischievous crime is a select subset of these crimes.

[33] Jacob and Lefgren estimate that school attendance decreases property crime by approximately 14%. They also estimate that violent crime increases on school days by about 28%. My estimate of the change in violent crime (31.5%) is approximately 10% larger than previous estimates, although my estimate is not statistically different from Jacob and

Table 4
Negative binomial regression onto total crime with suburban sub-sample

Variables	Suburban sub-sample				
	(I)	(II)	(III)	(IV)	(V)
Strike	−0.027	−0.064	−0.159	−0.162	−0.449
	(0.12)	(0.29)	(0.61)	(0.62)	(1.74)
Median income	−0.000015**	−0.000016**	−0.000013**	−0.000012**	–
	(8.01)**	(6.54)**	(4.74)**	(5.66)**	–
Welfare	−1.929**	−2.146**	−1.483**	−2.005**	–
	(9.21)**	(8.31)**	(4.20)**	(4.43)**	–
Poor parental educ.	1.639**	1.856**	1.683**	2.393**	–
	(13.33)**	(12.29)**	(8.09)**	(8.55)**	–
Juvenile maleness	−1.432**	−1.973**	−2.151**	−2.899**	–
	(5.33)**	(5.94)**	(4.74)**	(5.66)**	–
Student employment	−0.121*	−0.198**	−0.172	−0.064	–
	(2.23)*	(2.96)**	(1.87)	(0.59)	–
Single parent house	2.285**	2.369**	0.217**	3.062**	–
	(14.13)**	(11.97)**	(7.83)**	(8.82)**	–
Alpha	1.326**	1.303**	1.240**	1.203**	0.969**
	(59.08)**	(54.84)**	(39.01)**	(37.78)**	(34.42)**
Number of obs.	134,100	88,841	45,794	37,992	37,992
Time fixed effects	Y	Y	Y	Y	Y
Zip fixed effects	N	N	N	N	Y
R-squared	0.0294	0.0300	0.0302	0.0227	0.051

Note. Table 4 limits the sample to include only zip codes where 51% or more of the population lives in an area considered by the Census to be Suburban.

In columns I–V, the sample is re-specified (Type I through Type V) as in Table 2.

* Significant at the 5% level.

** Idem., 1% or smaller.

alcohol crimes, and weapons and endangerment crimes remain unaffected by school incapacitation.[34]

These changes in criminal activity seem to make sense. It seems logical that mischievous crime would be most affected by school incapacitation. After all, these are the types of crimes that often result from boredom rather than calculated criminal thought. On the other hand, the decrease in violent crime that occurs is harder to explain. I use, as Jacob and Lefgren do, social interaction theory to explain this result. Jacob and Lefgren argue that juvenile violence against other juveniles arises, at least in part, from disputes formed in and around the classroom. When these juveniles are not forced to be at school, there is a decrease in the amount of overall violent crime that reflects a decrease in juvenile violence against other juveniles. This explanation seems credible.[35] Given the limitations of the data set, I cannot test this theory further.[36]

Lefgren's. It is also true however that a mean comparison test (t test) of these point estimates shows that while my estimate of changes in property crime is statistically different from Jacob and Lefgren's ($t = 2.61$), the converse is not true. Therefore it is possible that all of these differences have occurred by chance.

[34] In total numbers, as opposed to percentage changes, these findings indicate that as many as 5 additional mischievous crimes, 6 additional property crimes and 4 fewer violent crimes may occur on strike day, with each one of these changes representing less than 0.03% of annual statewide crime.

[35] In 1998 and 1997, 62% of victims of juvenile violence were 17 or younger.

[36] I have no information concerning the victims of the crimes in my data set.

Table 5
Negative binomial regression onto total crime with rural sub-sample

Variables	Rural sub-sample				
	(I)	(II)	(III)	(IV)	(V)
Strike	0.017	0.027	−0.343	−0.285	−0.392
	(0.09)	(0.14)	(1.45)	(1.20)	(1.65)
Median income	−0.000034**	−0.000036**	−0.000036**	−0.000033**	–
	(26.68)**	(22.50)**	(16.62)**	(13.59)**	–
Welfare	−0.077	−0.063	−0.108	0.117	–
	(0.68)	(0.45)	(0.56)	(0.48)	–
Poor parental educ.	−0.282**	−0.332**	−0.218	−0.200	–
	(3.81)**	(3.63)**	(1.77)	(1.32)	–
Juvenile maleness	−0.357**	−0.362**	−0.370**	−0.421**	–
	(37.69)**	(32.83)**	(25.81)**	(25.64)**	–
Student employment	0.142**	0.156**	0.180**	0.267**	–
	(4.34)**	(3.86)**	(3.25)**	(4.15)**	–
Single parent house	0.379**	0.401**	0.317*	0.864**	–
	(4.41)**	(3.81)**	(2.15)*	(5.21)**	–
Alpha	2.679**	2.767**	2.810**	2.712**	2.197**
	(80.71)**	(66.36)**	(49.08)**	(46.97)**	(44.48)**
Number of obs.	1,101,359	729,469	377,244	304,491	304,491
Time fixed effects	Y	Y	Y	Y	Y
Zip fixed effects	N	N	N	N	Y
R-squared	0.0313	0.0308	0.0316	0.0218	0.0551

Note. Table 5 limits the sample to include only zip codes where 51% or more of the population lives in an area considered by the Census to be Rural.

In columns I–V, the sample is re-specified (Type I through Type V) as in Table 2.

* Significant at the 5% level.

** Idem., 1% or smaller.

The evidence so far seems to indicate that urban school districts are impacted by strike days in ways that do not necessarily translate to rural and suburban school districts. As a result I generally restrict my sample to urban school districts for further analysis. However it should also be noted that my choice of sample restriction does not significantly impact the main findings in the following sections, in particular that these results are only applicable to urban school districts.

4.3. Types of offenders

In addition to the topics already spoken to, the detailed level of the data also allows me to also explore the nature of the criminal, beyond superficial characteristics. By observing criminal record, I attempt to answer whether these additional crimes are caused by normal delinquents facing greater criminal opportunity, or by juveniles who do not normally engage in criminal activity. I again partition the dependent variable into two measures:

(1) total crime by repeat offenders, and
(2) total crime by one-time offenders.

I define repeat offenders not just as those juveniles who have committed previous crimes, but also those juveniles who will commit future crimes. Thus a juvenile who commits multiple crimes

Table 6
The effects of strike days on juvenile crime, by crime type and by community type

Dependent variable	Strike coefficient for		
	Urban sub-sample	Suburban sub-sample	Rural sub-sample
Drugs and alcohol	0.200	−1.188	−0.113
	(1.36)	(1.48)	(0.20)
Mischievous	**0.536****	−0.813	−0.241
	(3.88)**	(0.73)	(0.34)
Property	**0.257****	−0.128	−0.276
	(3.79)**	(0.34)	(0.80)
Violent	**−0.364***	−1.103	−0.266
	(2.50)*	(1.44)	(0.52)
Weapon and endangerment	−0.014	−53.096	–
	(0.05)	(0.02)	–
Number of obs.	129,593	37,992	304,491
Time fixed effects	Y	Y	Y
Zip fixed effects	Y	Y	Y
Ave. *R*-squared	0.0506	0.0423	0.0579

Notes. Each row represents a new regression with the listed crime type as the dependent variable. The reported number is the strike coefficient for the regression with the corresponding dependent variable. Each column is a different sub-sample group: urban, suburban and rural. All 14 regressions are defined as a Type V regression, including data from only April, September and October from 1984–2001, and zip code fixed effects. The Pseudo *R*-squared reported at the bottom of each column is the average *R*-squared over the 5 regressions. The marginal effects of these variables show that violent crime decreases by 31.53% on strike days. Property crime and mischievous crime increase by 28.81% and 48% respectively. Again, these effects reflect changes in juvenile crime as a proportion of the mean. The coefficients of the marginal effects are (−0.032), (0.072) and (0.025) for violent, property and mischievous crime respectively.

* Significance at the 5% level.

** Idem., 1% or smaller.

is considered to be a repeat offender juvenile even on their first crime. The results of these regressions can be seen in Table 7.

Additional overall crime is induced for both offender types; however, the average increase in crime for one-time offenders as a percent of the mean crime level for that group is much higher than for the repeat offenders. On strike days there is a 38.2% increase in average total crime for one-time offenders, and a 15.6% increase in average total crime for repeat offenders. Essentially, one-time offenders as a group seem to be much more affected by the lack of school incapacitation then the repeat offenders. This evidence suggests that the increased crime on strike days is less about criminal motive, and is more a result of boredom.

If these two groups are motivated differently to commit crimes, these effects may not be expressing differences across groups in the same types of crimes. As a society we may be more concerned if the lack of school incapacitation were inducing more property crime by one-time offenders rather than repeat offenders (where the crime may have eventually happened anyway). Therefore I am interested in examining whether these two groups differ in offense types on these strike days. To do this I create a dependent variable that equals total crime conditional on offender type and offense type. The results are expressed in Table 7. The decrease in violent crime comes mainly from the repeat offenders. Violent crimes committed by repeat offenders drop by an average of 36.7% on strike days. In addition, both groups are contributing to the overall increases in mischievous crimes and property crimes. Property crimes by one-time offenders and repeat offenders increase by an average of 56.7 and 25.4% respectively. Likewise mischievous crimes increase by 83.1% for one-time offenders and by 30.8% for repeat offenders.

Table 7
The effects of strike days on juvenile crime, by crime type and by offender type

Dependent variable	Strike coefficient for	
	One-time offenders	Repeat offenders
Total crime	**0.382****	**0.156****
	(4.05)**	**(2.77)****
Drugs and alcohol	0.337	0.153
	(1.18)	(0.93)
Mischievous	**1.024****	**0.597****
	(3.66)**	**(3.99)****
Property	**0.534****	**0.220****
	(4.44)**	**(2.81)****
Violent	−0.567	**−0.408****
	(1.48)	**(2.59)****
Weapon and endangerment	−1.330	−0.009
	(1.20)	(0.03)
Gateway crime	–	**0.286***
	–	**(2.00)***
Number of obs.	129,593	129,593
Time fixed effects	Y	Y
Zip fixed effects	Y	Y

Notes. Table 7 includes only urban zip codes. All 13 regressions are defined as a Type V regression, including data from only April, September and October from 1984 to 2001, and zip code fixed effects.The marginal effects of these variables show that for repeat offenders, violent crime decreases by 36.72%, property crime increases by 25.4%, and mischievous crime increases by 30.82% on strike days. The coefficients of these marginal effects are (−0.032), (0.046) and (0.012). For one-time offenders property crime increases by 56.75% and mischievous crime increases by 83.11% on strike days. The coefficients of these marginal effects are (0.030) and (0.008). Again, these effects reflect changes in juvenile crime as a proportion of the mean.

* Significance at the 5% level.
** Idem., 1% or smaller.

4.4. Strikes days as gateway crimes

Since the increase in crimes committed on strike days is substantially larger for first-time offenders, it may also be true that these juveniles may carry this behavior forward after their first arrest. To see if this is the case I examine whether or not a current day repeat offender was significantly more likely to haven gotten his/her start (commit their very first crime) on a strike day versus a normal school day. To do this, I generate a daily crime total that expresses the total of "first time" crimes by subsequent repeat offenders. The results of this regression can be also viewed in Table 7.

When using total first-time offenses by future repeat offenders as the dependent variable ('Gateway crime' on Table 7), the reported strike coefficient is positive and significant. This provides evidence that not only are more crimes being committed on strike days overall, but that crimes for future repeat offenders are also increasing on strike days relative to ordinary school days. This may suggest that school incapacitation is not only preventing the creation of crimes, but it may also be preventing the creation of criminals.[37]

[37] This finding by itself is not conclusive proof that strike days provide a gateway to future criminal action for those with no prior criminal history. Clearly more study is needed to establish such a relationship.

4.5. Displacement of crime

Thus far it seems the evidence suggests juvenile crime increases when students are not incapacitated in school. However what is not clear is whether these changes describe an overall increase in the amount of juvenile crime, or a displacement of crime from one day to another. Though Jacob and Lefgren try to speak to this question of temporal displacement in their paper, their use of in-service days again limits the power of their analysis. Jacob and Lefgren find that there is no temporal displacement of crime to in-service days; however, because in-service days are completely known, this result may simply be showing that juvenile criminals will plan to commit more crime when facing more criminal opportunity. What we are really interested in is whether a juvenile will temporally displace a "planned" crime given an unexpected opportunity to do so, or whether an unexpected opportunity to commit crime results in more "unplanned" crime. Again teacher strikes provide a more ideal measure of unexpected criminal opportunity, so I revisit this question of temporal displacement. Strike days give us a clearer picture as to whether school attendance is preventing the creation of new crime.

To test whether crime is being displaced, I begin by aggregating Total Crime to a weekly measurement rather than daily.[38] If juvenile crime is simply being displaced from weekend crimes to weekday crimes, then I should not observe an effect for the strike variable. To capture the weekly aggregated effect of strikes, I proportion the percent of the week where strikes occurred as a new measure of the strike treatment. Despite this aggregation, I find that strikes are still a significant factor in contributing to additional juvenile crime (see Table 8A).

Table 8A shows a positive and significant strike coefficient on 'Total crime' in the full data set. This is a strong indication that overall juvenile crime is increasing, rather than being displaced across time. However, if displacement of crime varies across crime type, or criminal type, we cannot conclude that no crimes are being displaced. To see if the type of crime or the characteristics of the perpetrator influences whether displacement of crime occurs, I partition weekly total crime by crime type and offender as before. The results of these regressions are also in Table 8A. These strike coefficients on Table 8A show that every specification of crime type and criminal type results in an increase of overall juvenile crime, and not temporal displacement of crime. I confirm these results by aggregating further to a monthly crime total. These results can be seen on Table 8B.

Weekly and Monthly aggregations of juvenile crime provide a good test of displacement of crime. I perform another test by lagging the strike effect, in order to pick up acute changes in crime shortly after the strike ends. If fewer crimes are committed on days following the strike, then the coefficient of the lagged strike variable should be negative and significant.[39] I run seven separate regressions, shown in Table 9, labeled "1 day" through "7 days." For a strike that ends on day t, the first treatment of the lagged strike variable under "1 day" begins on day $(t + 1)$, under "2 days" begins on day $(t + 2)$, and so on. Table 9 shows that there is no significant effect from the lagged strike treatment.[40] These results provide further evidence that juvenile crime occurring on strike days is additional crime, rather than displaced crime.

[38] This weekly measurement includes weekends and holidays, since strike day crime could displaced from usual weekend crime. In addition, week fixed effects are used in place of day fixed effects.

[39] This sample period also includes weekends. I also revert back to day fixed effects rather than week FE.

[40] I have also performed this test including the original strike day measurement with the lagged measurement in the regression. I find that this does not significantly alter the results.

Table 8A
Weekly crime by crime type and by offender type

Weekly aggregated dependent variable	Strike coefficient for			
	Full data set	Urban sub-sample	One-time offenders	Repeat offenders
Total crime	**0.235****	**0.243****	**0.418****	**0.176****
	(4.50)**	**(4.67)****	**(4.51)****	**(3.11)****
Mischievous	–	**0.570****	**0.820****	**0.501****
	–	**(4.39)****	**(3.09)****	**(3.55)****
Property	–	**0.312****	**0.492****	**0.225****
	–	**(4.73)****	**(4.27)****	**(3.06)****
Violent	–	**−0.300***	−0.223	**−0.317***
	–	**(2.32)***	(0.73)	**(2.27)***
Number of obs.	440,640	117,491	117,491	117,491
Time fixed effects	Y	Y	Y	Y
Zip fixed effects	Y	Y	Y	Y
Ave. psuedo *R*-squared	0.1119	0.1168	0.0564	0.1289

Table 8B
Monthly crime by crime type and by offender type

Monthly aggregated dependent variable	Strike coefficient for	
	Full data set	Urban sub-sample
Total crime	**0.534****	**0.485****
	(3.86)**	**(3.73)****
Number of obs.	100,980	26,946
Time fixed effects	Y	Y
Zip fixed effects	Y	Y
Pseudo *R*-squared	0.1134	0.1265

Notes. Each row in Table 8A represents a new regression with the listed crime type aggregated at the weekly level as the dependent variable. The reported number is the strike coefficient for the regression with the corresponding dependent variable. Each column is a different sample group. Column 1 is the full sample. The remaining three columns are from the urban sample. All 10 regressions are defined as a Type I regressions including zip code fixed effects. Table 8B shows 2 regressions, where the coefficients reported represent the strike effect on Monthly Total Crime.

* Significance at the 5% level.

** Idem., 1% or smaller.

4.6. Time trends of criminal activity

One last advantage I gain from using teacher strike days as my treatment for student absence, in conjunction with the strength of the daily arrest data, is that I can track how juvenile crime changes over time with prolonged absence. If it were true that an increase in juvenile crime is at least partially due boredom and inactivity on the part of juveniles, then I should expect that the longer juveniles remain "inactive," the more crime is likely to result. Arguably there is just as much uncertainty over when teacher strikes will end as when they will begin. As such, trends in juvenile crimes up until the time of the strike *and* over the period of the strike should not be biased by student or parent foresight. In addition, the longer a strike continues, the less able a parent or guardian is to provide adequate supervision to students. For example, a parent may be able to take one or two days away from work to supervise their child, but the cost of taking 10 consecutive days off work is substantially larger.

Table 9
Temporal displacement of crime through lagged strike variable

Lagged strike variable	Strike coefficient for urban sub-sample
1 Day	−0.033
	(0.62)
2 Days	−0.028
	(0.54)
3 Days	−0.030
	(0.57)
4 Days	−0.046
	(0.87)
5 Days	−0.041
	(0.78)
6 Days	−0.093
	(1.75)
7 Days	−0.070
	(1.33)
Number of obs.	293,112
Time fixed effects	Y
Zip fixed effects	Y
Ave. *R*-squared	0.0597

Notes. Each row represents a new regression with strike variable lagged the by the corresponding row. Each regression is defined as a Type IV regression including zip code fixed effects of the urban sub-sample. The notable exception is that weekends are included to allow for the correct number of strike treatments. Therefore, days where lagged strike treatments overlap breaks, holidays and other non-typical school days are included in the sample.

To examine if juvenile crime is changing with the length of a teacher strike I partition the strike variable into 4 groups of new independent variables. Three of these variables are clustered groups of 3 consecutive strike days, and the last variable retains the value of the strike for when a strike is in its 10th day or higher.[41] These results can be viewed in Table 10. Table 10 shows that for the first 3 days, there is no significant change in overall juvenile crime.[42] After the 5th day or so, the strike coefficient becomes positive and significant, reflecting an increase in overall crime. Further, after the 5th or so strike day there is a gradual increase in the level of juvenile crime as strike length continues.[43] It seems that as time passes, property and mischievous crime trends upward, while the decrease in violent crime remains relatively constant across days. The positive coefficient shows that these increases in property and mischievous crime overpower the decreases in violent crime. Column 2 of Table 10 repeats the regression in column 1, only with the strike variable clustered at the two-day level rather than at the three-day level. Clearly the evidence suggests that the longer juveniles remain absent from school, the more overall daily juvenile crime will be observed.[44]

[41] I do this because if I were to partition the strike variable for each separate day (1st day, 2nd day, etc.), then the number of treatments would be very small for the later strike days and the coefficients would be unreliable.

[42] When I break up the dependent variable (total crime) by crime type as before, violent, property and mischievous crime types are individually significant. However in the first 3 days, the opposite signs and magnitudes of the coefficients wash out aggregate effects in total crime. These results are consistent with Jacob and Lefgren's findings.

[43] Although the estimated effect of strike days on crime increases as strike length grows, these point estimates are not statistically different from one another.

[44] It is worth keeping in mind that these strike day counts are also limited to school days. That is to say that when a strike has reached its 10th day, a student has missed exactly 2 full weeks of school. It is also very likely that these

Table 10
Changes in total juvenile crime, by days elapsed since the start of a strike

Partitioned strike variable	Strike coefficient	Partitioned strike variable	Strike coefficient
Days 1–3	−0.005	Days 1–2	0.058
	(0.06)		(0.58)
Days 4–6	**0.185**	Days 3–4	0.006
	(1.91)		(0.07)
Days 7–9	**0.261***	Days 5–6	**0.262***
	(2.54)*		**(2.37)***
Day 10+	**0.313****	Days 7–8	**0.284****
	(2.93)**		**(2.62)****
Alpha	**0.533****	Days 9–10	0.228
	(101.05)**		(0.98)
–	–	Day 11+	**0.305****
			(2.67)**
–	–	Alpha	0.533**
			(101.04)**
Number of obs.	454,953		454,953
Time fixed effects	Y		Y
Zip fixed effects	Y		Y
R-squared	0.0713		0.0713

Notes. Table 10 includes only urban zip codes. These two regression are defined as a Type I regressions, using data including all ordinary school days from 1980 to 2001, and zip code fixed effects. Both regressions use Total Crime as their dependent variable.

* Significance at the 5% level.
** Idem., 1% or smaller.

5. Robustness checks

The results of the analysis up to this point seem to point to a very clear story about the relationship between juvenile crime and school incapacitation. In this section we perform some robustness checks to ensure that these results are not spurious. I begin by testing whether the strike variable is correlated with some unobservable characteristic, and that perhaps the results are picking up some alternate relationship. For instance, if a strike event is in fact very foreseeable, then one might speculate that the increase in juvenile crime predicted by the strike variable is really picking up an increased police presence on days that strike occur. That is to say that the results are not reflecting a true increase in juvenile crime, but that there are simply more police on the streets capturing a larger portion of the same total amount of juvenile crime.

If this were true, then one would expect that the number of days between an offense and an arrest would be shorter on strike days. The graphs in Fig. 1 show that this is not the case. When I consider only zip codes where strikes did occur, I see that the percent of quick arrests[45] is significantly higher on regular school days than on strike days. I explain this by suggesting that on regular school days, police have a better idea who the likely criminals are (repeat offenders),

criminal behaviors will change after a strike has gone on for an extensive amount of time. At that point, juvenile crime may level-off or even decrease substantially. Of course given the sparse and limited nature of extremely lengthy teacher strikes, an analysis of that nature is inherently difficult not to mention highly unreliable.

[45] I define these as arrests where the associated offense occurred less than 48 hours before the arrest.

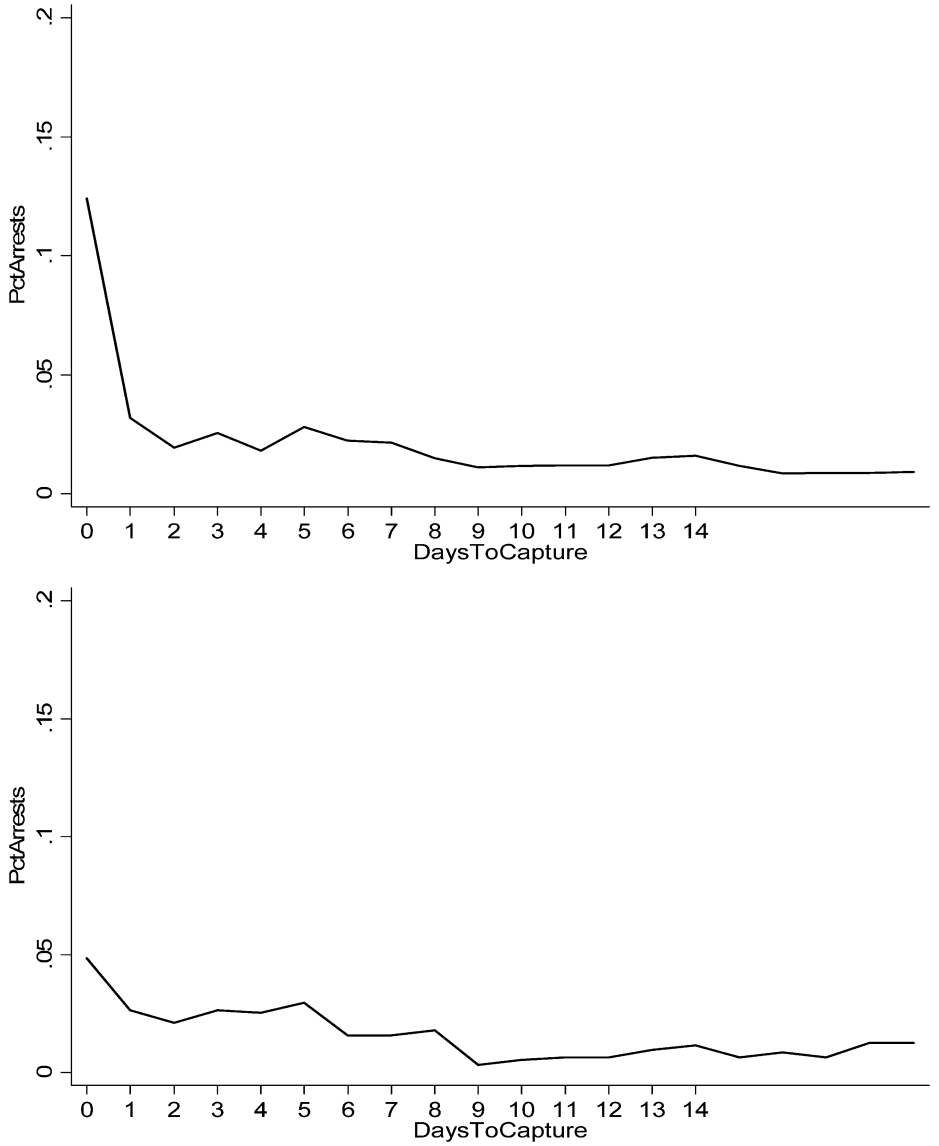


Fig. 1. Average percent of total arrests as function of days between offense and capture.

as opposed to strike days where offenders are more likely to be (previously unidentified) first time, or one-time offenders.

5.1. The importance of singular strikes

Given the uniqueness of strikes, I must confirm that these results are not being solely driven by one singular strike event. To do this I systematically eliminate each individual year from my sample starting with 1984. Table 11 shows that strikes are a significant determinant of juvenile crime in every year with the possible exception of 1985. When I drop 1985 from the sample

Table 11
The effect of strike days with specific years dropped from the sample

Year dropped	Strike coefficient	Year dropped	Strike coefficient
1984	**0.164****	1993	**0.166****
	(3.39)**		**(3.41)****
1985	0.093	1994	**0.163****
	(1.70)		**(3.28)****
1986	**0.166****	1995	**0.166****
	(3.43)**		**(3.40)****
1987	**0.160****	1996	**0.157****
	(3.16)**		**(3.23)****
1988	**0.163****	1997	**0.153****
	(3.37)**		**(3.16)****
1989	**0.148****	1998	**0.156****
	(3.02)**		**(3.23)****
1990	**0.189****	1999	**0.157****
	(3.77)**		**(3.26)****
1991	**0.229****	2000	**0.144****
	(3.18)**		**(2.99)****
1992	**0.157****	2001	**0.150****
	(3.24)**		**(3.11)****
Without Seattle	**0.133***	Seattle only	0.153
	(2.04)*		(1.82)
Ave. number of obs.	358,274		358,274
Time fixed effects	Y		Y
Zip fixed effects	Y		Y
Ave. *R*-squared	0.0603		0.0603

Notes. The top half of Table 11 is a regression of Strikes on Total Crime with each year systematically dropped from the sample. Each regression includes only urban zip codes. All 18 regressions are defined as a Type I regressions, using data including all ordinary school days from 1984 to 2001, and zip code fixed effects. The bottom half of the table reports the effect of strike days, for the same Type I sample, with Seattle dropped from the sample, and with Seattle making up the entire sample. The bottom of the table shows the average number of observations for the top 18 regressions, as well as the average *R*-squared term.

* Significant at the 5% level.
** Idem., 1% or smaller.

I find a much weaker significance than when I eliminate any other year. In September of 1985, the Seattle school district engaged in a teacher strike that lasted 20 school days. Given that the evidence has suggested that juvenile crime in urban districts is most affected by teacher strikes, it is possible that these results are being solely driven by the circumstances surrounding this one district.

To test this condition I restrict the sample twice, once to look at the sample without the Seattle school district, and then again to look at only the Seattle school district. If Seattle is driving all of the results, then the strike variable should be insignificant when I exclude Seattle, and extremely significant when I exclude all other districts except for Seattle. Table 11 shows that strikes still have a positive and significant effect when I drop Seattle from the analysis. In addition I see that the strike variable in the Seattle sub-sample is marginally significant at best, and does entire explain the results found in the urban sub-sample.[46] Therefore I can be confident that while this

[46] In fact, it is not more significant than the entire sample which includes districts that are unaffected by teacher strikes.

one event obviously contributes to the result (as I would expect the second longest strike in the most urban school district in Washington State to do), it is not the sole force driving the results.[47]

5.2. Reproducibility of results

Another robustness check I perform is to make sure that the results cannot be easily reproduced with random generations of the strike variable. To check if this is the case, I use a uniform random variable to re-specify a random dummy strike variable. Since teachers are on strike 2351 days out of the 1,703,910 day sample, strike days occur 0.13797% of the time. I assign a strike value of 1 to the random variable, when the value of the random observation generated is less than or equal to 0.0013797. If the randomly generated observation is greater than 0.0013797, then it is reassigned a value of zero. Since the random variable is uniform on the interval 0 to 1, the new random strike variable should approximate the number of strikes in the original sample. The results in Table 12A show the mean of the random strike coefficient over 25 trial regressions, as well as the average number of random strike days generated. Of the 25 trials, only 2 random strike variable coefficients were significant. One of the statistically significant coefficients was positive, while the other was negative.

5.3. Further tests of the data

One final test of the data is to make sure that my finding of new criminal participation on strike days is not a byproduct mislabeling offenders. Since the crime data ends in 2001, it could be that in the years leading up to 2001 many future career criminals are being mislabeled as one-time offenders due to the fact that future arrests past 2001 are unknown. To test this possibility, I drop the last 4 years of the data sample and repeat the regressions using offender type as the dependent variable. Table 12B shows that this mislabeling does not affect the results. The effect of a strike on repeat offenders is nearly identical for either specification.

6. Conclusion

The results presented in this paper provide a clearer picture concerning the relationship between schooling and urban juvenile crime. My main findings are the following:

(1) The effects of school incapacitation found in Jacob and Lefgren's study can potentially be much larger. The evidence suggests that property crime could fall by as much as 29%, almost twice as much as what is predicted in the existing literature. Likewise, violent juvenile crime increases by as little as 31.53%, and as much as 36.72%. These estimates of increased violent crime represent a 10–25% increase in previous estimates.
(2) I confirm that these changes in daily crime reflect changes in total crime, and not a displacement of crime from one day to the next, regardless of crime type and criminal type.
(3) Different types of juveniles are affected differently by school incapacitation. A significant proportion of the increase in the level of violent crime is caused by repeat juvenile offenders, while one-time offenders contribute more to the decreases in property crime and crimes of mischief.

[47] The Seattle strike of 1985 is actually tied with the Mukilteo strike of 1990 (at 20 days) for the second longest strike in Washington State history from 1980–2001.

Table 12A
The average effect of randomly generated strike days on total crime

Dependent variable: Total crime	Random generations of strike days
Ave. random strike coefficient	0.002 (0.05)
Average number of strike days	2355
Number of trials	25
Number of obs.	1,703,910
Time fixed effects	Y
Zip fixed effects	N
Ave. R-squared	0.0380

Table 12B
The effects of strike days by offender type with the last 4 years of the sample dropped

Dependent variable	Strike coefficient for	
	One-time offenders	Repeat offenders
Total crime	**0.382**** **(4.05)****	**0.156**** **(2.77)****
Total crime, without 1998–2001	**0.383**** **(4.08)****	**0.128*** **(2.31)***
Number of obs.	129,593	129,593
Time fixed effects	Y	Y
Zip fixed effects	Y	Y

Notes. Table 12A is the average strike coefficient over 25 trials using a randomly generated strike variable. It includes the full data set. Table 12B is a Type-V regression using just the urban sub-sample, and includes zip code fixed effects. Z-statistics are given in the parentheses for every table.

* Significance at the 5% level.
** Idem., 1% or smaller.

(4) A failure to incapacitate juveniles will result in significantly more crime by those juveniles who might not have otherwise engaged in criminal acts or who rarely engage in such acts. In fact, juveniles who become repeat offenders are more likely to have gotten their criminal start when incapacitation was expected but not implemented, as opposed to ordinary school days.

Incapacitation seems to have the greatest implications for new and seemingly preventable urban crime. I find that on strike days, new criminals are engaging in new criminal acts, and adding to the whole of criminal activity. Incapacitating these juveniles seems to be effective at preventing at-risk urban youths in engaging in new and risky behavior. I do not find that school incapacitation has any significant effect in suburban and rural communities. It may be true that characteristics that differentiate these community types from urban communities, as well as each other, may alter the premise of juvenile criminal behaviors.

If the evidence presented in this paper is accurate, then what we are learning is that how we manage children's time outside of school is very important. If parents are strained to provide adequate supervision for their children, juveniles who we might consider to be at-risk may be the

ones who are affected most, though at-risk juveniles are not the only ones affected. Families and businesses also bear the burden of increased juvenile delinquency. These results have profound implications for many urban school policies and programs. How school districts budget their students' days off is no longer a trivial matter. For example, school districts with long breaks (like in a traditional school calendar) may have very different juvenile criminal behaviors than a school district with frequent but shorter breaks (like in a year-round school calendar). Differences in how school calendars overlap within and between districts should have an effect on the nature of juvenile crime. The length of the school day or school year in a district, school district policies regarding student attendance requirements, and even the nature of after school programs within a district are all policy issues that determine more than just a child's educational outcome. It is my goal to continue to explore this data set and scrutinize these kinds of school policies, so that I can better understand the impact that these kinds of policies may have.

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Appendix A. Strike summary statistics

Total number of strike and strike treatments in the 22-year sample:
Total strikes: 102 (64)[48]
Total strike treatments: 2353 (1958)
Total strikes and strike treatments in each community:
Urban Total strikes: 74 (43)
Total strike treatments: 1858 (1616)
Suburban Total strikes: 31 (17)
Total strike treatments: 97 (67)
Rural Total strikes: 56 (32)
Total strike treatments: 398 (275)
Longest strike length: 25 days
Mean strike length: 6.7 days (Median is 5 days)

[48] The numbers in parenthesis represent the value in Type IV specifications.

Distribution of strike days		Total strike treatments by strike year	
1 day	321	1980	106
2 days	211	1981	24
3 days	202	1982	0
4 days	197	1983	74
5 days	188	1984	2
6 days	177	1985	567
7 days	179	1986	27
8 days	170	1987	87
9 days	58	1988	0
10 days	49	1989	0
11 days	45	1990	222
12 days	44	1991	1016
13 days	44	1992	9
14 days	39	1993	4
15 days	39	1994	90
16 days	39	1995	75
17 days	38	1996	0
18 days	38	1997	0
19 days	34	1998	18
20 days	34	1999	1
21 days	30	2000	0
22 days	3	2001	11
23 days	3		
24 days	3		
25 days	3		

Appendix B. Descriptive statistics of juvenile arrests

Total number of arrests in the 22-year sample: 401,864

Total arrests by race

White: 263,768 (65.64%)
Black: 31,121 (7.74%)
Other: 35,366 (8.80%)
Unknown or not reported/missing: 71,609 (17.82%)

Total arrests by age

1 year: 656 (0.16%)
2 years: 530 (0.13%)
3 years: 499 (0.12%)
4 years: 442 (0.11%)
5 years: 481 (0.12%)
6 years: 570 (0.14%)
7 years: 693 (0.17%)
8 years: 1077(0.27%)
9 years: 2189 (0.54%)
10 years: 3818 (0.95%)
11 years: 6971 (1.73%)
12 years: 15,883 (3.95%)
13 years: 35,976 (8.95%)
14 years: 61,937 (15.41%)
15 years: 84,861 (21.11%)
16 years: 93,922 (23.37%)
17 years: 91,359 (22.73%)

Total arrests by sex

Male: 279,465 (69.54%)
Female: 122,399 (30.46%)

Percent of "out of county" crimes committed[49]

Average across all zip codes: 11.87% (approximately 47,701)

Range for individual zip codes: [51.74–2.89%]

Appendix C. Regressions of the full data set excluding zip codes with less than 500 students

Variables	Full data set				
	(I)	(II)	(III)	(IV)	(V)
Strike	**0.187****	**0.177****	**0.200****	**0.246****	**0.212****
	(3.88)**	**(3.64)****	**(3.94)****	**(4.77)****	**(4.27)****
Median income	−0.00003**	−0.00003**	−0.00003**	−0.00003**	–
	(103.76)**	(85.14)**	(61.69)**	(54.36)**	–
Welfare	1.120**	1.149**	1.250**	1.572**	–
	(18.28)**	(14.20)**	(11.22)**	(11.09)**	–
Urban	0.110**	0.089**	0.071**	0.046**	–
	(22.30)**	(14.42)**	(8.38)**	(5.23)**	–
Poor parental educ.	0.503**	0.523**	0.570**	0.802**	–
	(14.02)**	(11.83)**	(9.34)**	(9.39)**	–
Juvenile maleness	0.266**	0.195**	0.0829	0.265**	–
	(5.66)**	(3.33)**	(1.01)	(2.66)**	–
Student employment	0.224**	0.228**	0.225**	0.192**	–
	(11.67)**	(9.61)**	(6.88)**	(5.07)**	–
Alpha	0.922**	0.941**	0.927**	0.889**	0.705**
	(153.67)**	(117.63)**	(92.70)**	(85.48)**	(76.58)**
Number of obs.	879,441	582,543	301,056	249,745	249,745
Time fixed effects	Y	Y	Y	Y	Y
Zip fixed effects	N	N	N	N	Y
Pseudo R-squared	0.0397	0.0396	0.040	0.0308	0.0556

Z-statistics are given in the parentheses for every table.

** Significant at the 1% level or smaller.

References

[1] G.S. Becker, N. Tomes, Human capital and the rise and fall of families, Journal of Labor Economics (The Family and the Distribution of Economic Rewards) 4 (2) (1986) S1–S39.

[2] G.S. Becker, N. Tomes, Child endowments and the quantity and quality of children (in Part II: Labor supply and the family), Journal of Political Economy (Essays in Labor Economics in Honor of H. Gregg Lewis) 84 (2) (1976) S143–S162.

[3] A.E. Cuellar, S. Markowitz, A.M. Libby, The relationships between mental health and substance abuse treatment and juvenile crime, Working paper No. 9952, NBER, 2003.

[4] I. Ehrlich, On the usefulness of controlling individuals: An economic analysis of rehabilitation, incapacitation and deterrence, American Economic Review 71 (1981) 307–322.

[5] I. Ehrlich, J.C. Gibbons, On the measurement of the deterrent effect of capital punishment and the theory of deterrence, Journal of Legal Studies 6 (1977) 35–50.

[6] R.B. Freeman, Why do so many young American men commit crimes and what might I do about it? Journal of Economic Perspectives 10 (1996) 25–42.

[7] R.B. Freeman, Crime and the employment of disadvantaged youths, in: G. Peterson, W. Vroman (Eds.), Urban Labor Markets and Job Opportunity, Urban Press Institute, Washington, DC, 1992, pp. 201–238.

[49] An "Out of County" crime is one where the crime has occurred in a county other than the county where the perpetrator's home zip code is a part of.

[8] E.L. Glaeser, B. Sacerdote, Crime and social interactions, Quarterly Journal of Economics 111 (1996) 507–548.
[9] J.T. Grogger, Local violence, educational attainment, and teacher pay, Working paper No. 6003, NBER, 1997.
[10] B. Jacob, L. Lefgren, Are idle hands the devil's workshop? Incapacitation, concentration and juvenile crime, American Economic Review 93 (2003) 1560–1577.
[11] S. Levitt, Juvenile crime and punishment, Journal of Political Economy 106 (1998) 1156–1185.
[12] L. Lochner, Education, work, and crime: A human capital approach, International Economic Review 45 (2004) 811–843.
[13] L. Lochner, Individual perceptions of the criminal justice system, Working paper No. 9474, NBER, 2003.
[14] N.H. Mocan, D.I. Rees, Economic conditions, deterrence and juvenile crime: Evidence from micro data, Working paper No. 7405, NBER, 1999.